

Junzheng Yang

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EDUCATION

University of Pennsylvania

Master of Applied Economics and Data Science

Philadelphia, PA

Aug. 2026 – Dec. 2027

University of California, San Diego

Bachelor of Science, Joint Major in Mathematics & Economics

La Jolla, CA

Sept. 2023 – Mar. 2026

Additional Study: Minors in Computer Science and Finance

Relevant Coursework: Econometrics, Probability & Statistics, Machine Learning, Numerical Linear Algebra, Data Structures

TECHNICAL & ANALYTICAL SKILLS

Programming & Tools: Python (pandas, NumPy, SciPy, matplotlib), R, SQL, MATLAB, Excel Solver

Financial & Investment Analysis: DCF valuation, financial-ratio analysis, factor analysis, fixed-income/yield-curve analysis, portfolio optimization, Black-Litterman

Statistical & Risk Analysis: Regression, GARCH, VaR, sensitivity/scenario analysis, Monte Carlo simulation, maximum likelihood estimation

PROFESSIONAL EXPERIENCE

Shaanxi International Trust Co., Ltd. | Quantitative Research Analyst Intern | Xi'an, China

Jul. 2025 – Sep. 2025

- Built an end-to-end research workflow for data acquisition, cleaning, modeling, evaluation, and forecasting, automating factor analysis and accelerating model iteration for investment research.
- Analyzed A-share and Hong Kong equities using fundamental, valuation, growth, and behavioral factors; applied scoring, regression, and machine-learning methods to develop a reusable multi-factor library.
- Supported sector-allocation analysis using Black-Litterman, combining macroeconomic data with bottom-up industry research and event-driven signals to update over/underweight views after market shocks.
- Applied DCF valuation, macro/technical indicators, and volatility-based position sizing to market-trend analysis, targeting 9.14% annualized volatility and a 2.61 Sharpe ratio; built Python monitoring for NAV and portfolio-limit compliance.
- Developed a Python risk-monitoring workflow to track NAV, trigger alerts at RMB 0.8, and test eight portfolio rules covering single-stock concentration, bond exposure, restricted securities, and minimum equity allocation.

WorldQuant | Quantitative Research Assistant | Remote

Aug. 2024 – Jul. 2025

- Developed and evaluated systematic investment strategies in Python and MATLAB, generating momentum signals and standardized backtests with transaction costs, overnight returns, and turnover calculations.
- Automated batch strategy loading and signal-matrix generation with normalization and missing-value handling, supporting faster iteration across multiple factor strategies.
- Improved 15-minute TWAP-to-daily-return backtesting efficiency by over 30% and built performance analytics for annualized return, Sharpe, Sortino, Calmar, maximum drawdown, holding periods, and capital allocation.
- Built equity-curve and drawdown visualizations and annual weighted-Sharpe analysis to compare strategy performance, risk, and stability across time periods.

Western Securities Co., Ltd. | Summer Analyst, Quantitative Risk Management | Xi'an, China

Jul. 2024 – Aug. 2024

- Analyzed market, credit, and liquidity risk for U.S. bond proprietary trading using VaR, sensitivity analysis, scenario simulations, historical yield curves, bootstrapping, and GARCH volatility estimates.
- Evaluated investment companies using credit ratings, financial ratios, macroeconomic indicators, and regulatory developments; synthesized findings into quantitative risk-assessment memoranda for investment decisions.
- Optimized U.S. Treasury and corporate-bond allocations with Markowitz mean-variance analysis and Excel Solver to improve risk-adjusted returns and manage tail risk.

SELECTED ANALYTICAL PROJECTS

Credit Risk Modeling & Variable Selection | Data Analyst | R

Mar. 2025 – Jun. 2025

- Cleaned and structured loan-application data; built logistic-regression models for approval/default analysis and quantified effects of income, loan amount, term, and CIBIL score on borrower outcomes.
- Applied backward stepwise selection to remove insignificant variables, reducing AIC from 2,100 to 1,896 and residual deviance to 1,886 while improving model parsimony and predictive performance.

Monte Carlo Simulation & Statistical Estimation | Quantitative Modeling Analyst | R

Mar. 2025 – Jun. 2025

- Implemented inverse-transform sampling and maximum-likelihood estimation in R; validated simulated moments against theoretical values and quantified estimator bias (≈ 0.021) and variance (≈ 0.042) across 1,000 simulations.
- Developed Wald confidence-interval testing using Fisher Information, achieving 95.2% empirical coverage and a 4.8% Type I error rate to validate statistical inference under repeated simulation.